

Chapter 10: Epidemiology of Non-Communicable Diseases (NCDs)

Intended Learning Objectives

By the end of this chapter, students should be able to:

- 1. Define non-communicable diseases and distinguish them from infectious diseases.**
 - 2. Describe the global and national burden of NCDs.**
 - 3. Identify major NCDs and their risk factors.**
 - 4. Explain the components of the Healthy Eating Plate and its role in disease prevention.**
 - 5. Recommend evidence-based dietary and physical activity guidelines.**
 - 6. Apply prevention strategies at individual and population levels.**
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1. Introduction

Non-communicable diseases (NCDs) are chronic conditions that are not spread from person to person. They develop over time, often silently, and are influenced by lifestyle, environment, and genetics. NCDs are the leading cause of death worldwide, responsible for over 70% of global mortality, and are increasingly prevalent in Egypt due to urbanization, dietary changes, and reduced physical activity.

2. Major Non-Communicable Diseases

Disease Group	Examples
Cardiovascular diseases	Hypertension, coronary artery disease, stroke
Cancers	Breast, lung, colorectal, cervical
Chronic respiratory diseases	Asthma, COPD
Diabetes mellitus	Type 1 and Type 2
Mental health disorders	Depression, anxiety, schizophrenia
Musculoskeletal disorders	Osteoarthritis, osteoporosis

3. Risk Factors

Modifiable Risk Factors

- **Tobacco use**
- **Unhealthy diet**
- **Physical inactivity**
- **Excessive alcohol consumption**
- **Obesity**
- **High blood pressure, high cholesterol, high blood sugar**

Non-Modifiable Risk Factors

- **Age**
 - **Gender**
 - **Genetic predisposition**
 - **Family history**
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4. The Healthy Eating Plate

The Healthy Eating Plate is a visual guide developed by nutrition experts to promote balanced, disease-preventing meals. It replaces outdated food pyramids with a clear, practical model.

Components of the Healthy Eating Plate

Section	Recommendations
Half of the plate	Vegetables and fruits – variety and color; limit potatoes and starchy vegetables
One quarter	Whole grains – brown rice, whole wheat, oats; avoid refined grains
One quarter	Healthy proteins – fish, poultry, legumes, nuts; limit red and processed meats
Healthy oils	Use olive, canola, sunflower oils; avoid trans fats and hydrogenated oils

Section	Recommendations
Water	Preferred drink; limit sugary beverages and fruit juices
Physical activity	Shown beside the plate to emphasize daily movement

5. Recommended Healthy Diet

Increase Consumption Of:

- Fruits and vegetables: ≥ 5 servings/day
- Whole grains: Brown rice, whole wheat bread, bulgur
- Legumes: Lentils, beans, chickpeas
- Healthy fats: Olive oil, nuts, seeds
- Low-fat dairy: Yogurt, laban, cheese

Limit Consumption Of:

- Added sugars: $<10\%$ of daily energy
 - Salt: $<5\text{g/day}$ (≈ 1 teaspoon)
 - Saturated and trans fats: Found in fried foods, margarine, pastries
 - Processed meats: Sausages, luncheon meats
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6. Recommended Physical Activity

WHO Guidelines for Adults (18–64 years)

- **150–300 minutes/week of moderate-intensity aerobic activity**
- **OR 75–150 minutes/week of vigorous-intensity activity**
- **Muscle-strengthening activities ≥ 2 days/week**
- **Reduce sedentary time; break long sitting periods**

Children and Adolescents (5–17 years)

- **≥ 60 minutes/day of moderate to vigorous activity**
- **Include aerobic and bone-strengthening exercises**

Older Adults (65+ years)

- **Same as adults, with added focus on balance and flexibility**
- **Activities should be adapted to functional ability**

7. Prevention and Control Strategies

Individual-Level

- **Lifestyle modification: diet, exercise, smoking cessation**
- **Routine screening: blood pressure, glucose, BMI**
- **Medication adherence and chronic disease management**

Population-Level

Strategy	Examples
Health education	School and media campaigns
Policy and regulation	Tobacco taxation, food labeling, salt reduction
Environmental changes	Walkable cities, clean air initiatives
Health system strengthening	NCD clinics, universal coverage, integrated care

✔ Multiple Choice Questions (MCQs)

1. Which component should occupy half of the Healthy Eating Plate?

- A. Whole grains**
- B. Fruits and vegetables**
- C. Proteins**
- D. Dairy products**

Correct Answer: B. Fruits and vegetables

2. Which of the following is a recommended source of healthy fat?

- A. Margarine**
- B. Butter**
- C. Olive oil**
- D. Palm oil**

Correct Answer: C. Olive oil

3. WHO recommends how many minutes of moderate-intensity physical activity per week for adults?

- A. 60 minutes**
- B. 150–300 minutes**
- C. 500 minutes**
- D. 30 minutes only**

Correct Answer: B. 150–300 minutes

4. Which of the following is a modifiable risk factor for NCDs?

- A. Age**
- B. Gender**
- C. Family history**
- D. Physical inactivity**

Correct Answer: D. Physical inactivity

5. Which dietary pattern is most associated with cardiovascular protection?

- A. High-fat ketogenic diet**
- B. Mediterranean diet**
- C. Western fast-food diet**
- D. Low-protein diet**

Correct Answer: B. Mediterranean diet